Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2025-26

Compiler Construction Lab Batch 2022-26

Lab Assignment No: - 5	
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Batch	2022-26
Class	CSE – B3
Academic Year & Semester	4th year, 7th Semester
Date of Performance	
Title of Assignment	Conversion of decimal to hexadecimal number in a file.
Practice Questions	1. Write a LEX program for conversion of decimal to hexadecimal number in a file. 2. Write a LEX program for decimal to binary conversion.

Source Code	1. %{ #include <stdio.h> int r,digit;char a[26]; %} %% [0-9]+ { int num=0;int count=0;int len=yyleng; for(int i=0;i<len;i++){ num=num*10+(yytext[i]-'0'); }

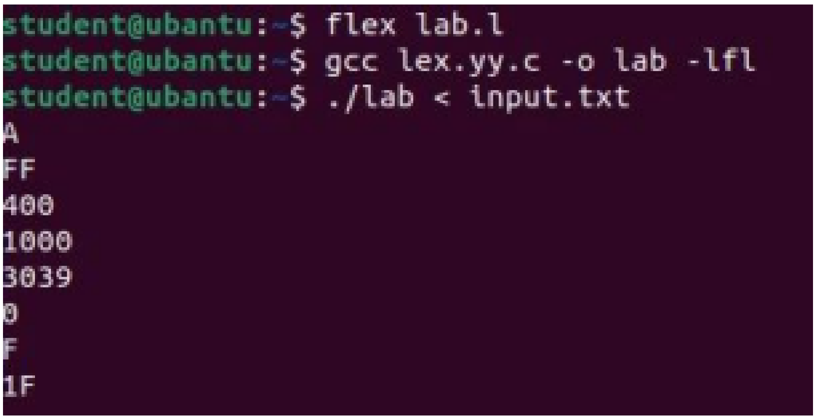
	`` ```

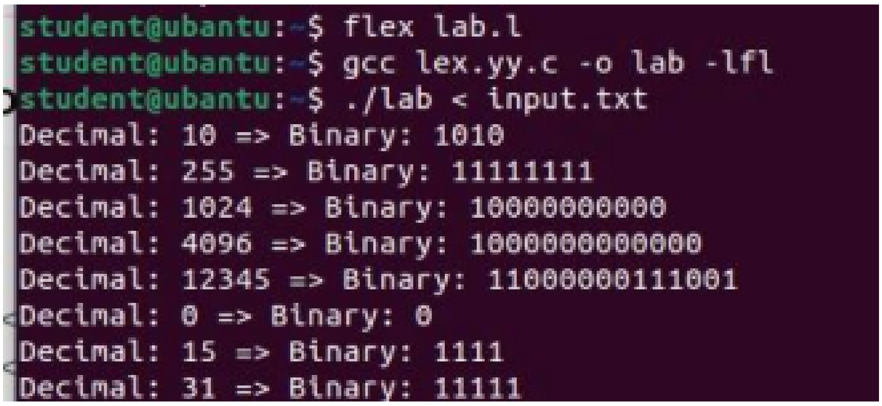
```
if(num==0){
printf("0");
}
else{
while(num!=0){
r=num%16;
if(r<10){
digit='0'+r;
}
else{
digit='A'+(r-10);
}
a[count++]=digit;
num=num/16;
}
for(int i=count-1;i>=0;--i){
printf("%c",a[i]);
}
}
}
. {printf("%c",yytext[0]);}
%%
int main(){
yylex();
return 0;
}
int yywrap() {
return 1;
}
```

2.

```
%{
#include <stdio.h>
#include <stdlib.h>

void decimalToBinary(int n) {
 if (n == 0) {
 printf("0\n");
 return;
 }

 int binary[32];
 int i = 0;

 while (n > 0) {
```

	binary[i++] = n % 2; n /= 2;

	} // Print in reverse order for (int j = i - 1; j >= 0; j--) printf("%d", binary[j]); printf("\n"); } %} %% [0-9]+ { int decimal = atoi(yytext); printf("Decimal: %d => Binary: ", decimal); decimalToBinary(decimal); } [\t\n]+ ; // ignore whitespace . ; // ignore other chars %% int main(int argc, char *argv[]) { yylex(); return 0; }
Output Screenshot	1.  2.



Post lab questions

1. LEX code for case conversion of alphabets using file handling. CODE:

```
%{
#include <stdio.h>
#include <stdlib.h>
%}

%%
[0-9A-Fa-f]+ {
 // Convert hex string to decimal integer
 int decimal = (int)strtol(yytext, NULL, 16);
 printf("Hexadecimal: %s => Decimal: %d\n", yytext,
decimal); }
[ \t\n]+ ; // ignore whitespace
. ; // ignore other chars
%%

int main() {
 yylex();
 return 0;
}
```

OUTPUT:

	student@ubantu:~$ flex lab.l student@ubantu:~$ gcc lex.yy.c -o lab -lfl student@ubantu:~$ ./lab < input.txt Hexadecimal: A => Decimal: 10 Hexadecimal: FF => Decimal: 255 Hexadecimal: 400 => Decimal: 1024 Hexadecimal: 1000 => Decimal: 4096 Hexadecimal: 3039 => Decimal: 12345 Hexadecimal: 0 => Decimal: 0 Hexadecimal: F => Decimal: 15 Hexadecimal: 1F => Decimal: 31
Conclusion	We have successfully implemented a lex program to convert decimal numbers to hexadecimal numbers.